Brazilian E-Commerce

Exploratory Data Analysis & Business Insights

99,441 orders · 5 tables · Olist dataset

Python | Pandas | NumPy | Seaborn

Habiba Anwer· Data Analyst Portfolio

Business context

A Brazilian online marketplace (Olist) shared its real order data.

Questions I set out to answer

1. How many orders actually reach the customer?
2. What does a typical order look like?
3. Which categories and products drive sales?
4. Where are the customers?
5. Is shipping cost related to price?

Data understanding

99,441	112,650	32,951
orders	order items	products

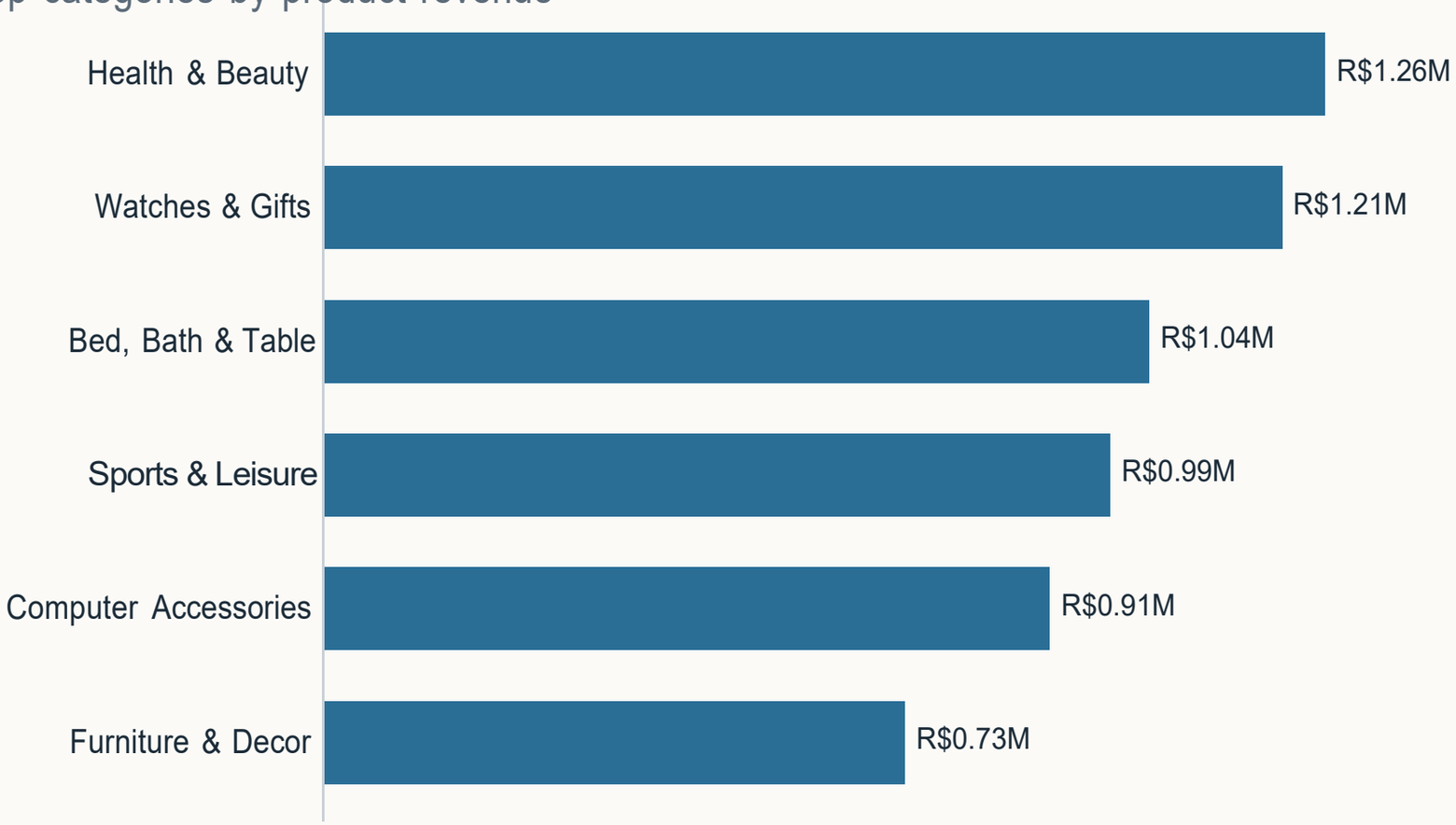
99,441	103,886
customer records	payment records

Quality checks

- No duplicate rows in any table
- Missing delivery dates belong to orders that never reached that stage: kept
- 610 products have no category: left as missing, excluded from category figures
- Dates converted to datetime; raw data kept untouched in working copies

Product & revenue insights

Top categories by product revenue



Volume ≠ revenue

Bed, bath & table sells the most units (11,115), but health & beauty earns the most. One product sold 527 times; another made R$48.9K from just 35 sales.

Revenue = sum of item prices (no freight, not profit)

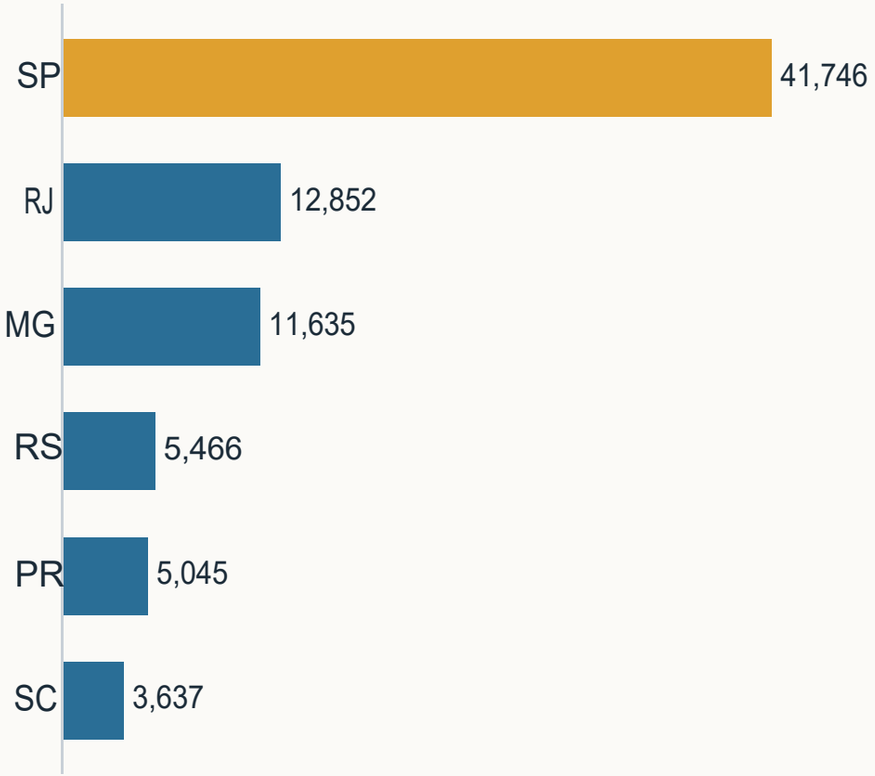
Customer & order insights

97.02%
orders delivered
(96,478 of 99,441)

R$137.75
average order value

R$86.90
median order value

Orders by customer state (top 6)



São Paulo alone = ~42% of all orders

The mean sits above the median: a few very large orders (max R$13,440) pull the average up.

Delivery & shipping

R$22.82
average freight per order

R$160.58
average order total including shipping

0.41
price–freight correlation

What it means

- Shipping adds a meaningful cost on top of the item price.

- Pricier items tend to cost more to ship, but the link is only moderate: price alone does not explain freight.

- 97.02% of orders reached the customer; 625 were canceled.

Delivery-time vs. estimate analysis is included in the notebook.

Key findings & next steps

Health & beauty leads revenue (~R$1.26M)

Followed by watches & gifts (R$1.21M) and bed, bath & table (R$1.04M).

Popularity and revenue differ

Best-selling products are not always the top earners.

Demand is concentrated

SP, RJ and MG account for about 67% of orders.

Averages hide skew

Median order (R$86.90) is well below the mean (R$137.75).

Ideas worth testing

→ Promote high-revenue, low-volume products differently

→ Compare delivery time and freight in less-served states

→ Check monthly seasonality before planning campaigns

Project & skills

Tools

Python · Pandas · NumPy · Matplotlib · Seaborn · Jupyter

Skills shown

Data cleaning & validation · Aggregation & merging
Descriptive statistics (percentiles, IQR, correlation)
Data visualisation · Business storytelling

Full notebook & README on GitHub

Link in the post

Thank you! Feedback is very welcome.